

DEAL DOA: L2

L1-L2 Strategy Session

TO & TO SMA for Sao Paulo Operations

MRS (Brazil)

South America

SFDC#232976

July 27th, 2026

Deal Team

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Project Management: Jay Evans & Nelyo Oliveira

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MRS (Brazil)

TO and TO SMA for Sao Paulo Operations Executive Summary – L1-L2 Strategy Session

07 27 2026

SFDC #	232976	Financial Items	Proposal Terms	Commercial Milestone(s)	Target
Type:	Budgetary	Currency	USD	Standard Wabtec T&Cs.	Proposal Need Date
CP 1 Complete - Product	NO	Payment Terms	Net 30		AUG26
CP 1 Complete- Services	NO	Cash Flow	TBD		AUG26
Expected Contract	1Q 2027	Incoterms	TBD		SEP26
Term	1+5 Years	Escalation	Y		Finalize Contract / Terms
Sales	M. Chamone	Down Payment	YES	Contract Terms	Contract Execution / PO
P&L Rep	K. Viegas/T. Duque/ E. V			To be negotiated	Q1'2027
CommOps	E. Garcia				Estimated Delivery
ContractMgmt	D. Oswald				Acceptance/Warranty Start
					12 Mos
					12 Mos

Deal Overview

Opportunity: Provide TO (perpetual licenses) for 51 AC44i Locomotives

- Offer Trip Optimizer solution (including GoLINC and Installation Kits)
- Include TO SMA (5 Years) adding value with: optimize TO utilization rate, local technical assistance, rail network updates (track database), SW bug fixes (equivalent to Rumo's offering)

Customer CTQs

- Train Handling: Operations standardization among different trains and operators (in-train forces more restricted than AAR) Growing for new segments at Rumo's tracks
- Fuel Saving: Enabler for tangible NPV savings
- Roadmap to Automation: Mid-Long plans for unmanned/ remote operations

Competitive Landscape:

- Progress Rail is visiting MRS and RUMO... fleet equipped make it harder for competitor's entry
- Doing nothing

TO SMA scope (same as Rumo):

- Software Fixes and Updates
- Project Engineering Support
- Track Database Changes
- TO Back Office Infrastructure
- Service Program Management
- TO Training
- Extended Hardware Warranty and Spare Material

2016 (Heavy Haul operations) Contractual Scope

- Customized solution for TO software at MRS segment (Heavy Haul/ Iron Ore Service) 69 SW & HW @ \$87,000/unit ExW Erie, US, subject to proven fuel savings = **\$6M**
- Full integration w CBTC keeping MRS accountable for ICD availability for integration
- Hold MRS safety standards
- Compatibility w/ 134 & 268 rail car trains (as well potential changes on train configuration)
- > = 2.65% fuel saving at 134 rail car train
- Technical support, training for train operators
- 10 years SW bug fixes for systemic failures
- 12 months warranty

Lessons Learned from 2016 Contract

Installed Base/ Assessment:

- Underestimated MRS in-train forces being more restricted than AAR...software changes
- Underestimated comm loss inside tunnels
- Underestimated AC44i Locomotive hasn't exactly the same configuration between MRS & USA Fleet
- Underestimated CBTC with TO Integration (GE & Wabtec as 2 separate companies)

Preliminary Financials and Strategy

Preliminary Financial Summary

QTY	Trip Optimizer	Extended Cost	Unit Price	Extended Price
51	TO SW License		\$ 83,287	\$ 4,247,637
51	GoLINC - 8 slot		\$ 26,600	\$ 1,356,600
51	Hardware	\$ 612,000		
	NRE	\$ 618,700		
	Track Mapping and deployment	\$ 115,000		
	PM (10 hrs/m 12 months)	\$ 15,360		
	Other Costs	\$ 479,912		
		\$1,840,972	Revenue	\$ 5,604,237
			Margin	\$ 3,763,265
			CM%	67%

Annual SMA: ~\$4M
~\$350K Cost
~\$850K Price (~65%CM)

Value Proposition

- Standardization on train handling under constant changes on train make-up and temporary speed restrictions
- Fuel savings
- Reduce training process

Deal Close Strategy

- Offer TO solution to earn credibility back on this discussion with MRS (after heavy haul service 'bitter' experiences)
- Offer alternative pricing options (out of USA & **Brazil supply**)
- Include fuel savings commitment based on opportunities for higher efficiencies
- Offer TO ZZZ functionality as phased approach

Risk and Opportunities:

- Risk : MRS will ask for fuel savings guarantee
- Oppty: MRS to include Heavy Haul operation (iron-ore carrousel) on follow-on order

Pricing

- Previous contract is the baseline for MRS
- SW/license price is OK
- HW price will be a negative surprise

Historical/Benchmark Pricing

2016 CONTRACTUAL PRICING

\$ 62.000	License
\$ 11.900	Go LINC with HPEAP
\$ 8.900	SCIO Upgrade
\$ 4.200	Installation Kit

\$ 87.000 Trip Optimizer Kit (2016)

Customer Expectation:

2026 SIMULATED CONTRATUAL PRICING

\$ 83.287	License
\$ 17.286	Go LINC with HPEAP
\$ 12.928	SCIO Upgrade
\$ 6.101	Installation Kit

\$ 119.602 Trip Optimizer Kit (2026)



Accelerating the future of transportation

Backup Slides

Lessons Learned from 2016 Contract

Installed Base/ Assessment

- Underestimated MRS in-train forces being more restricted than AAR...software changes
- Underestimated comm loss inside tunnels
- Underestimated AC44i Locomotive hasn't exactly the same configuration between MRS & USA Fleet
- Underestimated CBTC with TO Integration (GE & Wabtec as 2 separate companies)

Management

- No achievement on fuel savings: Underestimated MRS expertise on heavy haul service and their discipline for improving YoY their optimized operations
- GE Engineering resources limitation with 3 TO projects signed at the same time (MRS, Vale & Rumo)

MRS Locomotive Configuration



- AC44i (or ES44) Locomotives:
 - AC propulsion.
 - 6 axles CC.
 - Wide gauge.
 - Traction Power – 4400 THP.
 - STE/CTE – 200k lbf / 166k lbf.
 - CCA control architecture
 - These locomotives are very similar to the ES44AC locomotives, except they use 7FDL-LC diesel engines instead of the EVO engine.
 - Trip Init with PTC-Wabtec
- In the past few years (2018/2019) MRS ran a pilot with 4 Locomotives with TO installed for iron ore application. No tests were ever performed at MRS general cargo flow service (Rumo's tracks).
- Status of the Trip Optimizer installation is unknown.
 - TO has been disabled for several years
- TO software used in MRS in 2018/2019
 - CCA = 55.36.20
 - CCA Configs = various (all released)
 - TO Package = 37.03.05.01.204031 (i.e., 20.40.31)
 - EM Configs = EM000051, EM000052
 - HPEAP: RMD_APP_0421_560A.tgz

Loco 7331 and 7314 are already operating in RUMO tracks

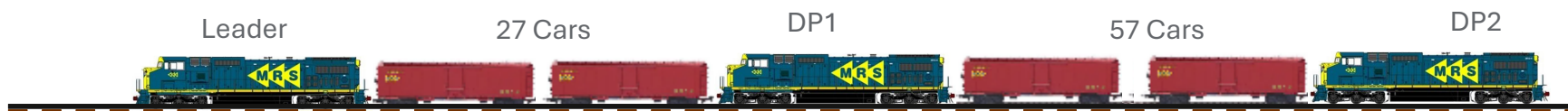
MRS's Train Information

- **Train Type:** Loaded and Empty Grain
- **Car Type:** Hopper Covered



- **Locomotives:** AC44 or ES44

Train Consist						
# Cars	Loco Position	DTM	Loaded Weight (Metric Ton)	Length (m)	Total Car Axles	Similarities with RUMO over 1 year
84	(1x27x1x57x1)	N	10920	1530	336	0.4%
120	(2x40x1x80x1)	N	15600	2250	480	3.1%
135	(2x43x1x92x1)	N	17200	2400	540	5.2%
135	(2x42x1x42x1x51)	N	17200	2400	540	Included in the 5.2% above

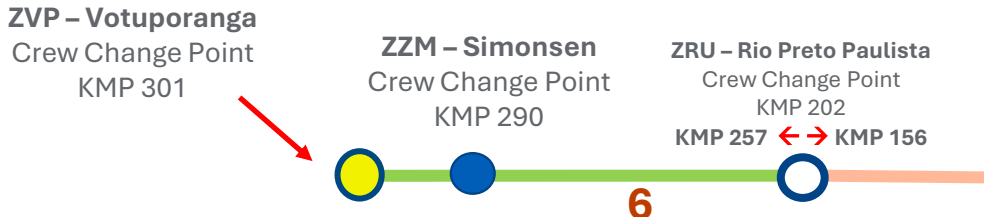
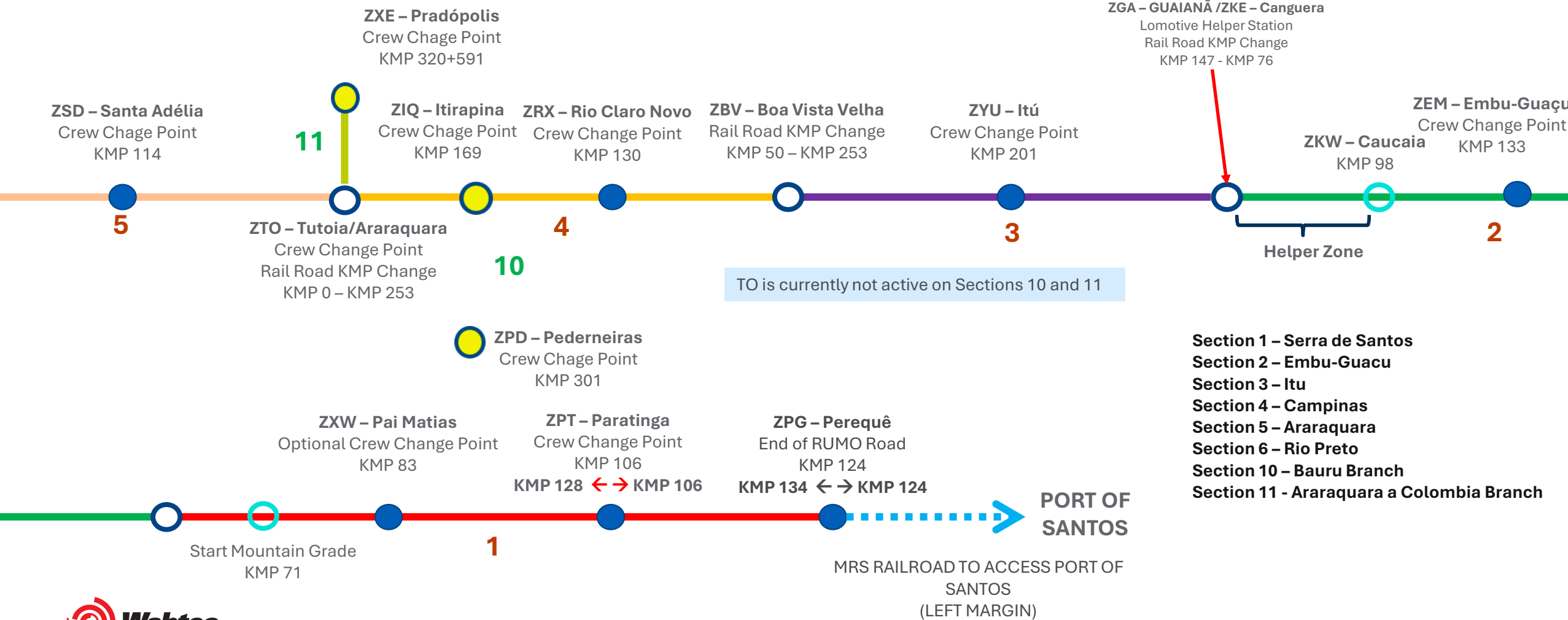


Train Consist Schema for 84 cars

MRS's Simple Track

-  MRS's Crew Change Point
-  RUMO's Crew Change Point
-  RUMO's Region Boundary

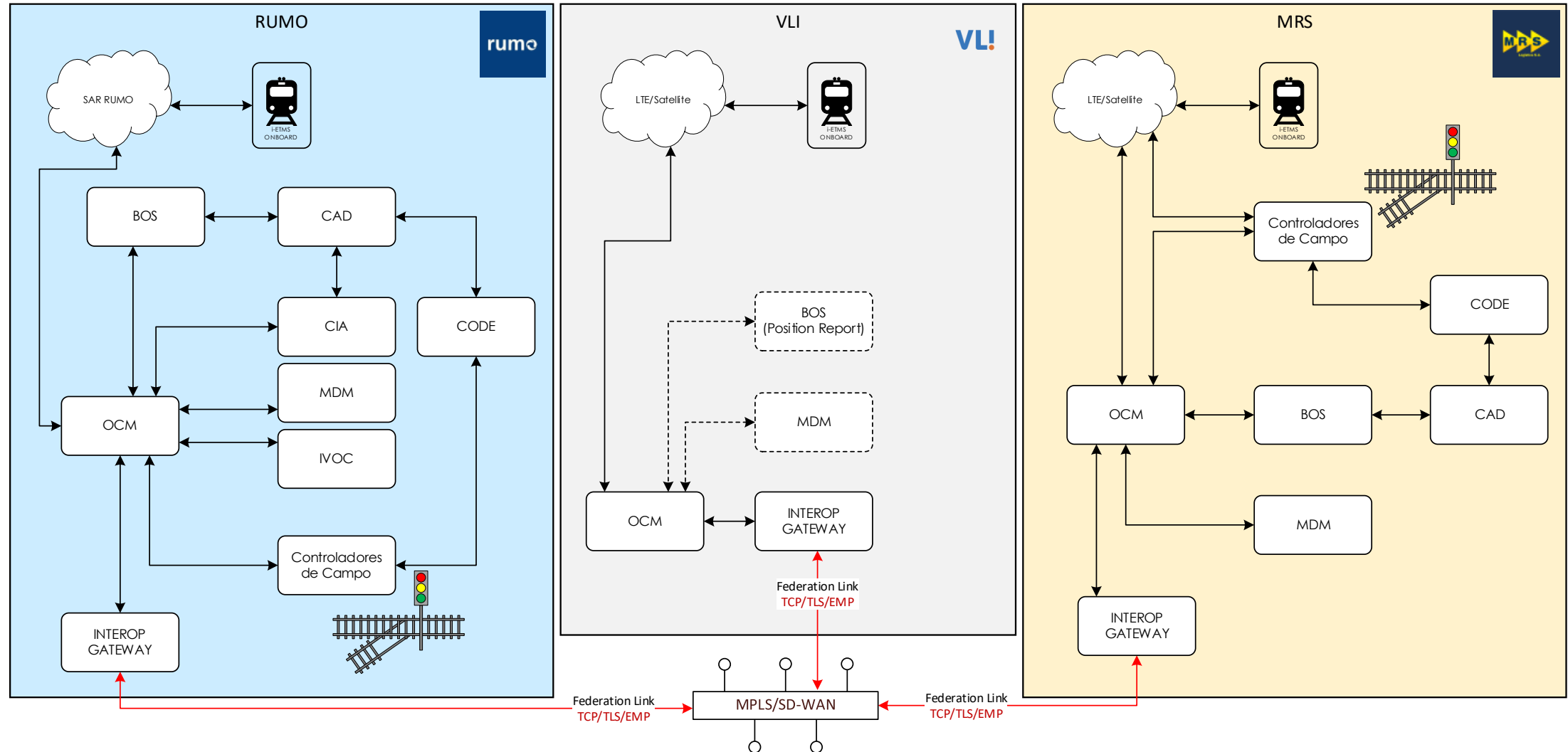
Origin	Origin ID	Destination	Destination ID
Pereque	ZPG	Votuporanga	ZVP
Itirapina	ZIQ	Pederneiras	ZPD
Tutoia	ZTO	Pradópolis	ZXE



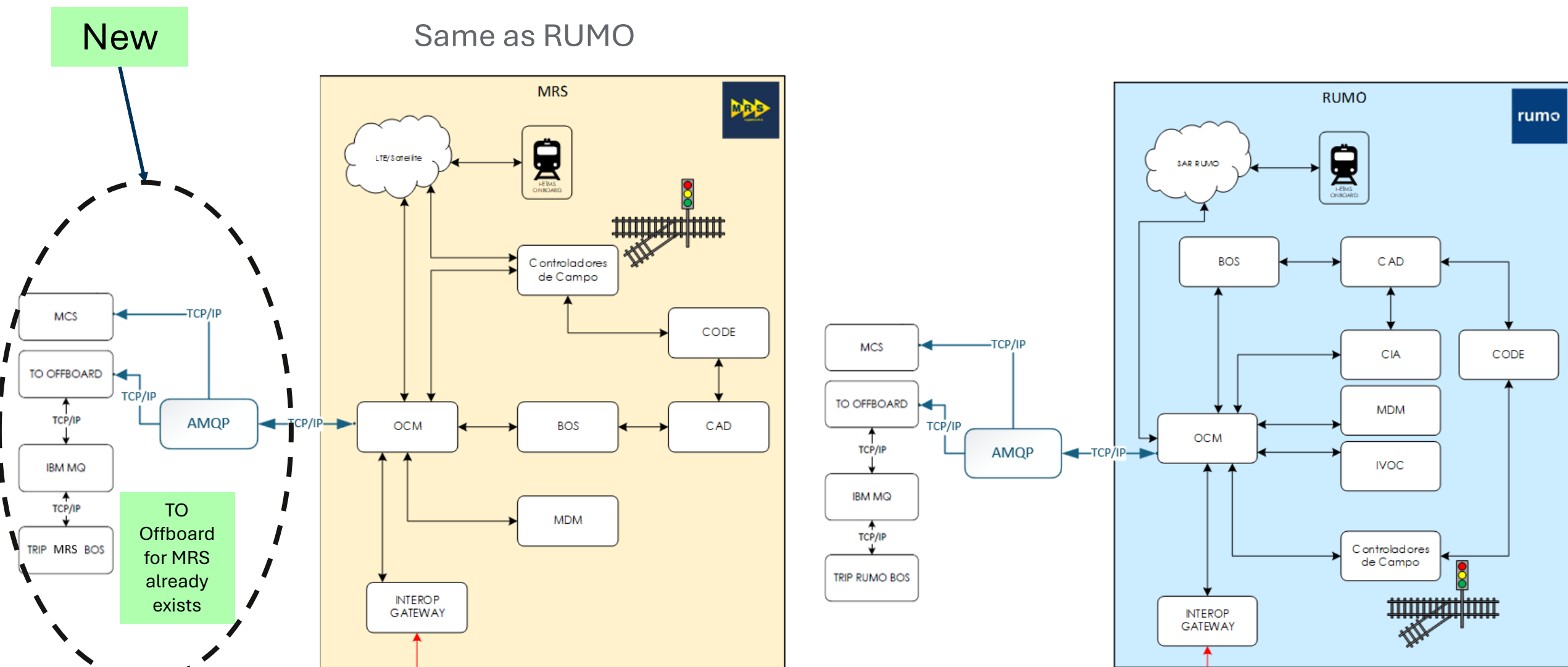
- Section 1 - Serra de Santos
- Section 2 - Embu-Guacu
- Section 3 - Itu
- Section 4 - Campinas
- Section 5 - Araraquara
- Section 6 - Rio Preto
- Section 10 - Bauru Branch
- Section 11 - Araraquara a Colombia Branch



Current PTC Proposed Architecture (Interoperability)



TO Proposed Architecture



MRS PTC-IF Interoperable Operation Proposal

3 Step approach

Step 1: Initial Simulation

Wabtec simulates operating conditions (no field visit required).

Objective: First view of in-train forces for 135-car trains.

Trigger: Agreement on in-train forces metric for Paulista Network.

Note: MRS is looking for values below 100 Kips, but not necessarily the same thresholds of the iron ore loop.

Step 2: Preparation for Qualification Test

Define boundary conditions for the project on MRS trains.

Develop document and methodology for the test.

Evaluate and adapt locomotives for the test.

Trigger: Methodology document approved + locomotives ready with Trip Optimizer active.

Note: Wabtec will effectively release the finalized software for MRS. Once approved, SW will be deployed on all locos.

Step 3: Multi-Year Agreement

Continuous improvement and joint operation between MRS and Wabtec for all locomotives on the Paulista Network.

Ensure Wabtec support for all lead locomotives.

Prepare MRS for future expansions (territory, operational models, technologies).

Trigger: Qualification Test approved + commercial/technical conditions defined + agreement signed.

Owning RR: MRS
Employing RR: MRS
Operating RR: RUMO

Premises:

No additional feature needed
(to be confirmed after
simulations).

Need to re-enable TO on pilot
locos (service work scope)

Need to review installation
process for TO on a Card
PTC Protect is fully deployed
and functional on MRS locos

Trip Optimizer Historical Pricing (2016–2026)

2016 Contract Pricing
Ex-Works USA

2026 Customer Price Simulation
Ex-Works USA

Simulated Contractual
Escalation

2026 SIMULATED CONTRATUAL PRICING		
\$	83.287	License
\$	17.286	Go LINC with HPEAP
\$	12.928	SCIO Upgrade
\$	6.101	Installation Kit
\$	119.602	Trip Optimizer Kit (2026)

2016 Business Case:
Customer pay back based on
2.65% fuel savings over 1M liters
burned per locomotive

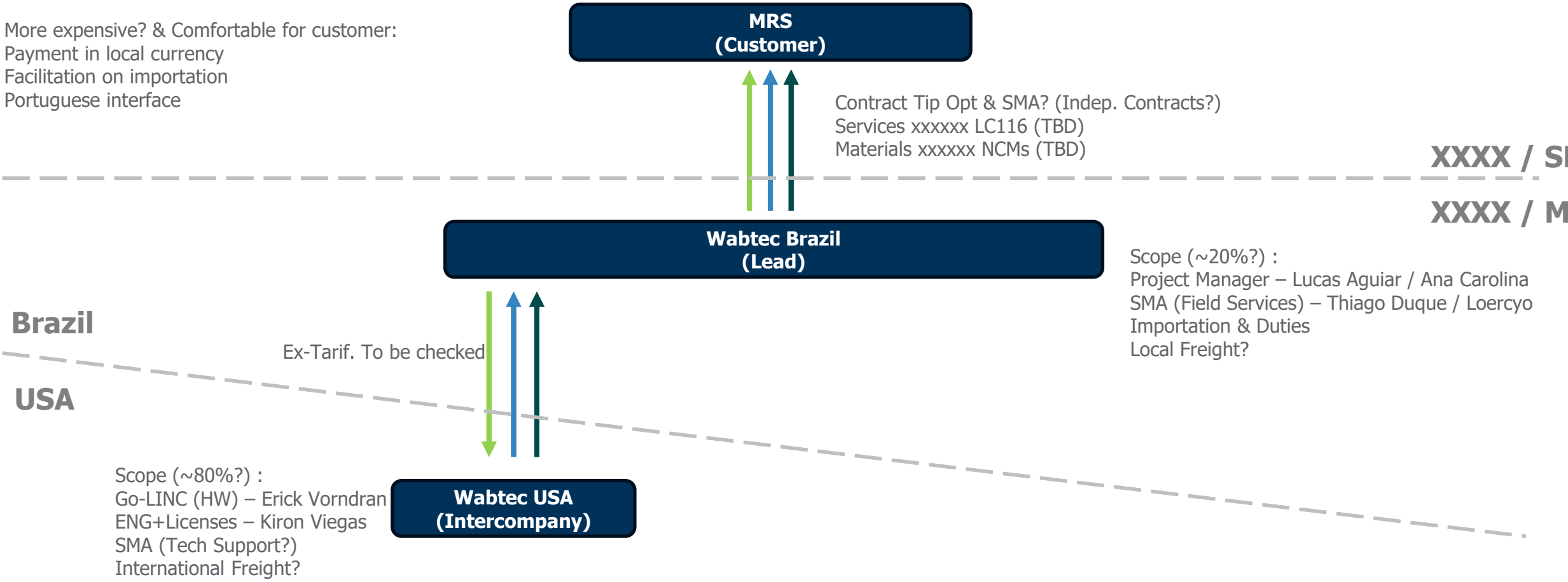
Key Takeaway: Customer value perception is primarily concentrated on the License and Installation Kit components.

/// COMMERCIAL SETUP – MRS – TO FOR SP

Scenario 1 – TO & SMA through Brazil



More expensive? & Comfortable for customer:
Payment in local currency
Facilitation on importation
Portuguese interface



/// COMMERCIAL SETUP – MRS – TO FOR SP



Scenario 2 – TO through USA // SMA through Brazil

Best price setup ?
Customer responsible for Imp&Duties
2 legal entities to deal with



Contract - Trip Opt
Services xxxxxx LC116 (TBD)
Materials xxxxxx NCMs (TBD)

Contract - SMA
Services xxxxxx LC116 (TBD)

EXW Incoterm

XXXX / SP
XXXX / MO



Scope (~15%?) :
Project Manager – Lucas Aguiar / Ana Carolina
SMA (Field Services) – Thiago Duque / Loercyo

Brazil

Ex-Tarif. To be checked

USA



Scope (~85%?) :
Project Manager - ?
Go-LINC (HW) – Erick Vorndran
ENG+Licenses – Kiron Viegas
Field Services - ?

/// COMMERCIAL SETUP – MRS – TO FOR SP

Scenario 3 – TO & SMA through USA

Good price setup ?
Portuguese interface
Customer responsible for Imp&Duties

**MRS
(Customer)**

Contract - Trip Opt & SMA
Services xxxxxx LC116 (TBD)
Materials xxxxxx NCMs (TBD)

PM - Portuguese Interface

EXW Incoterm

XXXX / SP

XXXX / MO

**Wabtec Brazil
(Intercompany)**

Scope (~7%?) :
Project Manager – Lucas Aguiar / Ana Carolina

Brazil

Ex-Tarif. To be checked

USA

Scope (~93%?) :
Project Manager - ?
Go-LINC (HW) – Erick Vorndran
ENG+Licenses – Kiron Viegas
Field Services - ?

**Wabtec USA
(Lead)**

2 - Confidential

→ Services / Licenses
→ Materials
→ Invoice